

Reframing Human-AI Collaboration for Generating Free-Text Explanations

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Abstract

Large language models are increasingly capable of generating fluent-appearing text with relatively little task-specific supervision. But can these models accurately explain classification decisions? We consider the task of generating free-text explanations using human-written examples in a few-shot manner. We find that (1) authoring higher quality prompts results in higher quality generations; and (2) surprisingly, in a head-to-head comparison, crowdworkers often prefer explanations generated by GPT-3 to crowdsourced explanations in existing datasets. Our human studies also show, however, that while models often produce factual, grammatical, and sufficient explanations, they have room to improve along axes such as providing novel information and supporting the label. We create a pipeline that combines GPT-3 with a supervised filter that incorporates binary acceptability judgments from humans in the loop. Despite the intrinsic subjectivity of acceptability judgments, we demonstrate that acceptability is partially correlated with various fine-grained attributes of explanations. Our approach is able to consistently filter GPT-3-generated explanations deemed acceptable by humans.

1 Introduction

As natural language understanding tasks have become increasingly complex, the field of explainable NLP has embraced explanations written in free-form natural language. In contrast to extractive explanations that highlight tokens in the input, free-text explanations provide a natural interface between machine computation and human end-users (Hendricks et al., 2016; Camburu et al., 2018). The dominant paradigm for producing free-text explanations is via direct supervision, i.e., training an autoregressive, generative language model to predict human-authored explanations directly (Kim et al., 2018; Park et al., 2018; Ehsan et al., 2018; Narang et al., 2020; Wiegreffe et al., 2021, i.a.).

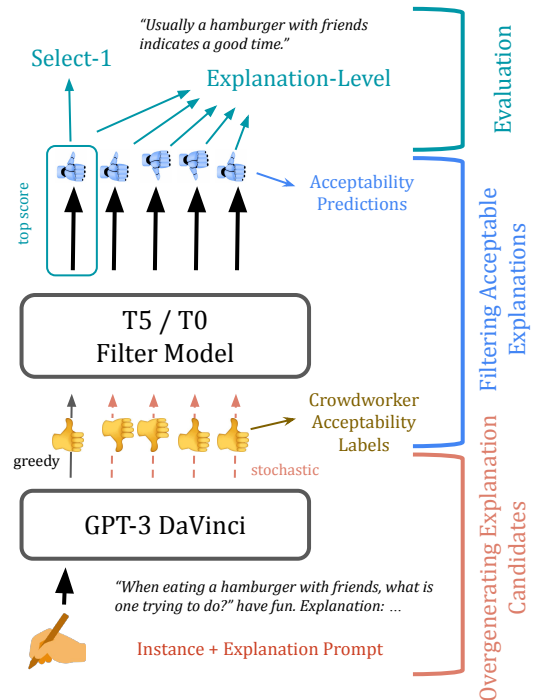


Figure 1: Illustration of our overgeneration + filtration pipeline for producing human acceptable explanations for CommonsenseQA and SNLI (see examples in Table 1). Authors of this work write explanations to prompt GPT-3, generating 5 explanations per instance. An acceptability filter, trained on human binary acceptability judgments, determines which of these generated explanations are plausible. Evaluation is performed at both the explanation and the instance level.

However, collecting high-quality written explanations to serve as supervision is difficult and expensive. More than 70% of existing free-text explanation datasets are crowdsourced (Wiegreffe and Marasović, 2021), and even the most meticulous crowdsourcing efforts frequently fail to elicit logically consistent and grammatical explanations (Narang et al., 2020). Furthermore, a lack of standardized crowdsourcing design has resulted in highly varied datasets, which are hard to compare or combine (Tan, 2021).

Recent progress in prompting large language models (LLMs) provides a potentially promising alternate to large-scale crowdsourcing. The *in-context learning* paradigm, wherein powerful language models are prompted in a few-shot manner with just a few examples, has proven surprisingly effective across a range of NLP tasks (Radford et al., 2019; Brown et al., 2020; Shin et al., 2020; Schick and Schütze, 2021a, *i.a.*). In this work we ask: can LLMs also generate reliable *explanations*? In human subjects studies, we find that GPT-3 (Brown et al., 2020) can be readily made to generate explanations via prompting, and surprisingly, humans often prefer GPT-3 generated explanations to crowdsourced explanations in existing datasets (§2).

Two additional human subjects studies, however, demonstrate that GPT-3-generated explanations still have significant room for improvement along axes such as providing new information (i.e., avoiding repetition) and supporting the label; human subjects found less than half of greedily-decoded GPT-3-generated explanations to be *acceptable* with 100% agreement. To improve upon this, we *re-frame* the role of crowd annotators: instead of asking them to write explanations as in prior work, we (1) repeatedly query GPT-3 to generate multiple candidate explanations for each input instance, and (2) ask crowdworkers to rate the acceptability of each candidate generation. After showing that GPT-3 can usually generate an explanation that humans unanimously find acceptable within as few as five queries (§3), we use a small number of these binary crowdworker judgments to supervise an acceptability filtering model, which can be applied to select high quality candidates among GPT-3’s outputs (Figure 1; §4).

Despite intrinsic subjectivity in acceptability ratings, our supervised model improves upon the already-competitive few-shot paradigm by consistently selecting (human-identified) high quality explanations better than strong baselines. Human evaluations reveal that the filtration model not only improves acceptability, but also other axes like supporting the label and providing novel information.

In summary, our main findings are:

- i. few-shot prompting with GPT-3 produces surprisingly competitive explanations, providing a promising alternative to crowd-authored free-text explanation corpora;
- ii. binary human labeling can instead be leveraged

- to train a filter that selects high-quality machine-generated explanations; and
- iii. in areas where GPT-3 struggles, including information content, supporting the label, and overall acceptability, our proposed overgenerate-and-filter pipeline improves generated explanations. We publicly release our code and data.¹

2 GPT-3 is Competitive with Crowdsourced Explanation Datasets

We investigate three research questions:

1. Are GPT-3-generated explanations preferable to crowdsourced ones in existing datasets? (§2.1)
2. Can improving prompt quality improve GPT-3-generated explanations? (§2.2)
3. Along what fine-grained dimensions are GPT-3-generated explanations preferred, and do these correlate with overall acceptability? (§2.3)

Explanation tasks and datasets. We consider two English tasks: CommonsenseQA and natural language inference (NLI), shown in Table 1. CommonsenseQA (Talmor et al., 2019) is a multiple choice task posed over commonsense questions. Crowdsourced free-text explanations for instances in CommonsenseQA are provided in the CoS-E v1.11 (Rajani et al., 2019) and ECQA (Aggarwal et al., 2021) datasets. ECQA explanations are counterfactual, i.e., annotators were instructed to explain not only the correct answer choice but also why the others are incorrect. ECQA was released to address the quality issues of CoS-E (Narang et al., 2020); for completeness, we experiment with both. Our second task is NLI, which involves inferring whether a given hypothesis sentence entails, contradicts, or is neutral towards a premise. This task is instantiated with the SNLI dataset (Bowman et al., 2015) and crowdsourced explanations from e-SNLI (Camburu et al., 2018). For each task, we report results on a fixed, randomly-sampled 250-instance test set not observed during prompt design.

Few-shot prompting for explanations. We use GPT-3 Davinci² (Brown et al., 2020), an autoregressive language model with ~175B parameters trained on a large dataset of text scraped from the internet. We prompt the model with several (question, gold answer, explanation) triplets followed by an unexplained question-gold answer instance for

¹https://github.com/allenai/few_shot_explanations/

²Non-instruct version, available in Summer 2021.

SNLI (Bowman et al., 2015)
Premise: Dark-haired man wearing a watch and oven mitt about to cook some meat in the kitchen.
Hypothesis: A man is cooking something to eat.
Label: entailment
e-SNLI (Camburu et al., 2018): Meat is cooked in a kitchen, and is a food that you eat. Using an oven mitt implies you’re about to cook with hot utensils.
GPT-3: Cooking is usually done to prepare food to eat.
CommonsenseQA (Talmor et al., 2019)
Question: What is the result of applying for job?
Answer Choices: anxiety and fear, increased workload, praise, less sleep, or being employed
Correct Choice: being employed
CoS-E (Rajani et al., 2019): being employed applying for job
ECQA (Aggarwal et al., 2021): Applying for a job is followed by attending interview which results in being employed. Applying for a job may not result in the other options.
GPT-3: Applying for a job can result in being employed, which is a positive outcome.

Table 1: Task-specific instances, along with their crowdsourced explanations from the respective datasets, shown alongside explanations generated greedily by GPT-3. In our experiments, the SNLI GPT-3 explanation was preferred over its corresponding e-SNLI explanation by 2/3 annotators. For CommonsenseQA, 3/3 preferred the GPT-3 explanation to the CoS-E explanation, and 2/3 to the ECQA one.

which we expect the model to generate an explanation.³ We use a total of 115 randomly sampled train instances to create our prompts; each prompt consists of 8-24 randomly selected examples from this set. For each instance, we generate a single explanation with greedy decoding. More details about prompt construction are in [Appendix A](#); example prompts are given in [Tables 2 and 11](#).

Crowdsourcing evaluation. Given that existing automatic metrics often do not correlate well with human judgements of explanation quality ([Clinciu et al., 2021](#); [Kayser et al., 2021](#)), we conduct human

³We condition on the gold label as a *methodological control* to ensure reliable human evaluation. In pilot studies, we found it hard to avoid bias against explanations when we disagreed with the predicted label. Prior work ([Kayser et al., 2021](#); [Marasović et al., 2021](#)) has removed this confounder by only considering explanations for *correctly-predicted* instances, which may overestimate explanation quality. Our method allows us to report results on a truly representative sample. We experiment with incorrect vs. correct predictions in [Appendix C](#), finding that GPT-3 can competitively explain gold labels for instances it predicted incorrectly.

Let’s explain classification decisions.
A young boy wearing a tank-top is climbing a tree.
question: A boy was showing off for a girl.
true, false, or neither? neither
why? A boy might climb a tree to show off for a girl, but he also might do it for fun or for other reasons.
###
A person on a horse jumps over a broken down airplane.
question: A person is outdoors, on a horse.
true, false, or neither? true
why? Horse riding is an activity almost always done outdoors. Additionally, a plane is a large object and is most likely to be found outdoors.
###
There is a red truck behind the horses.
question: The horses are becoming suspicious of my apples.
true, false, or neither? false
why? The presence of a red truck does not imply there are apples, nor does it imply the horses are suspicious.
###
A dog carries an object in the snow.
question: A dog is asleep in its dog house.
true, false, or neither? false
why?

Table 2: Example of a prompt with 3 training examples for SNLI: presented are the **premise/hypothesis** pairs, the **gold labels**, and the **explanations** (written by us) that act as input to GPT-3 (in practice, we use 8-24 examples per prompt). The text generated by the model acts as the free-text explanation. In this case, the model greedily auto-completes (given 12 examples): “A dog cannot carry something while asleep”.

studies for evaluation.⁴ We ensure each experiment has a substantial number of distinct crowdworkers to mitigate individual annotator bias ([Table 17](#)).

We present workers with a dataset instance, its gold label, and two explanations for the instance generated under different conditions (“head-to-head”). We then ask them to make a preferential selection, collecting 3 annotations per data point. We report inter-annotator agreement (IAA) using Krippendorff’s α ([Krippendorff, 2011](#)). We find low-to-moderate agreement across studies, indicating the subjective nature of the task; see [Tables 3, 4, and 5](#). [Appendix B](#) contains further details on quality control.

⁴Our studies were conducted on the Amazon Mechanical Turk platform, at the Allen Institute for AI. We selected workers located in Australia, Canada, New Zealand, the UK, or the US, with a past HIT approval rate of >98% and >5000 HITs approved, and compensated them at a rate of \$15/hour. Each worker completed qualifying exams on explanation evaluation and the NLI task.

Dataset	Preferred Explanation (%)			α
	Crowd	Tie	GPT-3	
CoS-E	20.3	34.9	44.8	0.5
ECQA	52.7	12.9	34.4	0.2
e-SNLI	63.6	11.6	24.8	0.3

Table 3: Head-to-head human evaluation for 250 explanations generated by GPT-3 vs. written by crowdworkers in the datasets, along with Krippendorff’s α for IAA. Results are shown as % preferences. We prompted GPT-3 with crowd explanations *from the corresponding datasets*.

2.1 Are GPT-3 explanations preferred over crowdsourced ones?

We perform a head-to-head comparison of explanations generated by GPT-3 with greedy decoding vs. gold human-written explanations in the original datasets. The crowdsourced explanations serve as a reasonable upper bound for what a supervised explanation generation model trained on them could produce. Table 1 contains examples of GPT-3-preferred explanations.

Results are shown in Table 3. GPT-3 greedily-decoded explanations are frequently preferred or comparable to crowdsourced explanations in CoS-E, which is not too surprising given the dataset has many ungrammatical explanations (Narang et al., 2020). And, while ECQA and e-SNLI explanations are strongly preferred to GPT-3, there are still a non-trivial number of cases where GPT-3 explanations are competitive (47.3% and 36.4%, respectively).

2.2 Can improving prompt quality improve GPT-3-generated explanations?

Dataset	Preferred GPT-3 Explanation (%)			α
	Crowd Prompts	Tie	Our Prompts	
CoS-E	6.9	13.5	79.6	0.2
ECQA	15.9	9.5	74.7	0.3
e-SNLI	30.8	26.8	42.4	0.5

Table 4: Head-to-head human evaluation for 250 explanations generated by GPT-3 prompted with either author-written explanations or crowdsourced explanations from the associated datasets, along with Krippendorff’s α for IAA.

Given that low-quality training instances may result in low-quality predictions (especially in a few shot setting),⁵ we ask: can we improve GPT-3 gen-

⁵For example, GPT-3 reproduces known data artifacts in the CoS-E corpus when prompted with explanations from it, such as the recurring phrase “rivers flow trough (sp) valleys”.

Dataset	Preferred Explanation (%)			α
	Crowd	Tie	GPT-3	
CoS-E	7.2	13.9	78.9	0.5
ECQA	44.5	9.7	45.7	0.4
e-SNLI	49.6	9.7	40.7	0.2

Table 5: Head-to-head human evaluation for 250 explanations generated by GPT-3 vs. written by crowdworkers in the datasets, along with Krippendorff’s α for IAA. GPT-3 explanations were prompted with explanations *written by the authors*.

erations simply by conditioning on higher-quality instances? To this end, we replace the 115 crowdsourced explanations from the original datasets for prompting GPT-3 with explanations carefully written by the authors of this paper (see Table 12 for examples). Our prompts are used to generate a different set of GPT-3 explanations on the same test data.

We perform a head-to-head human evaluation of the GPT-3 generations conditioned on the explanations we wrote vs. those conditioned on the crowdsourced explanations. Results in Table 4 show that, for all three corpora, generations conditioned on our explanations outperform generations conditioned on crowdsourced ones, illustrating the importance of good-quality prompts for GPT-3.

We repeat the experiment of §2.1, but with our prompts instead of dataset prompts. With this change, GPT-3 generations are even more competitive (Table 5). For all three datasets, more than half the time, few-shot prompting results in an explanation *at least as good as* a human-written explanation. For subsequent experiments, we prompt GPT-3 with the author-written explanations.

2.3 What types of explanations does GPT-3 generate?

Pairwise evaluations can only offer perspective on the relative quality of generated explanations. Are crowd annotators simply comparing explanations on surface-level features like grammaticality?

To understand finer-grained characteristics of explanations, we design a second human study to collect absolute Likert-scale judgments across seven axes of quality (with each explanation judged by 3 annotators). The first three axes capture surface-level features: generality, grammaticality, and factuality. The next three capture richer aspects of explanation quality: whether new information is introduced (a requirement for non-vacuous explana-

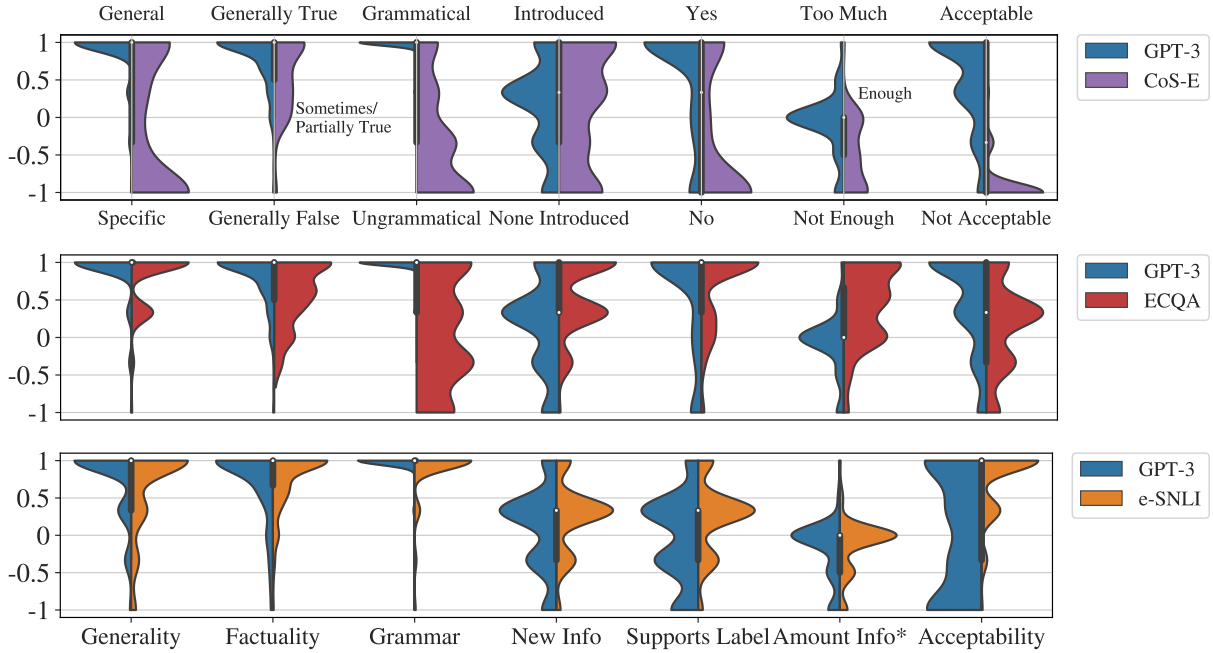


Figure 2: Absolute evaluation for GPT-3 and crowdsourced explanations for CommonsenseQA via CoS-E (top) and ECQA (middle) datasets, and for NLI via the e-SNLI dataset (bottom). The distribution of mean scores of 3 annotators for each instance in the test set is plotted. All attributes besides **Factuality** and **Amount Info** are binary. ***Amount Info** is the only attribute for which a value of 0 is preferred to a value of 1. See Table 18 for more details.

tions), whether explanations support the gold label, and whether the amount of information given is sufficient. Finally, we ask for an overall judgement of quality: *is the explanation acceptable?* We explain our design process in Appendix B.3. Results on the crowdsourced and GPT-3 explanations for both tasks are given in Figure 2.⁶

For both tasks, GPT-3 explanations do well in all 3 surface-level categories, with statistically significantly greater ratings in generality and grammaticality (and factuality for CommonsenseQA) compared to crowdsourced explanations, and distributional means close to 1. In these categories, there is little room for improvement.

On the other hand, GPT-3 explanations do not contain as much new information as ECQA and e-SNLI explanations, indicating substantial room for improvement (mean=0.1 for both tasks compared to 0.6 for ECQA and 0.2 for SNLI; these differences are statistically significant at $p \leq 0.01$). GPT-3 explanations are substantially more supportive of the label vs. CoS-E, but not as supportive as ECQA or e-SNLI (all statistically significant at $p \leq 0.1$). Indeed, the mean rating of GPT-3 explanations for label support is 0.5 for Common-

senseQA and -0.1 for NLI, demonstrating room for improvement. These axes are crucial to ensuring explanations are not vacuous and are on-topic. Finally, GPT-3 explanations are judged as acceptable at higher rates than CoS-E or ECQA explanations, but not e-SNLI explanations. Mean scores of 0.5 (CommonsenseQA) and 0.0 (NLI) indicate that GPT-3 explanations have room to improve overall.⁷

Correlation between acceptability and other attributes

To understand what factors are important for the overall “acceptability” judgement, we compute Spearman correlation (ρ ; Spearman, 1987) between acceptability and all other attributes (Table 6). Each is positively correlated with acceptability, though with varying degrees of magnitude. Acceptability is least correlated with “new information,” and most correlated with grammaticality, generality, and the explanation’s support for the label. Overall, the results indicate that human preference for explanations is not fully explained by any one attribute, and is not limited to surface-level features.

⁶Krippendorff’s α is 0.48 for CommonsenseQA annotations and 0.31 for SNLI; see Table 15.

⁷We do not evaluate explanations counterfactually, which may explain why ECQA explanations are often labeled as having “too much information”. See Appendix B.1.

Attribute	ρ	n
Generality	0.25	3750
Factuality	0.17	2958
Grammar	0.31	3750
New Info	0.05	3750
Supports Label	0.22	2943
*Amount Info	0.16	1761

Table 6: Spearman’s correlation (ρ) between acceptability and other attributes in Figure 2. For *Amount Info*, a value of 0 is preferred to a value of 1. All results are statistically significant at $p < 0.01$.

3 Beyond Greedy Explanations

While GPT-3 explanations demonstrate strength across surface-level features and are surprisingly competitive in head-to-head settings, they can still be improved. One might imagine a setting with multiple end-users in which we wish to provide the most **unambiguously acceptable** explanation as output from a system. When considering the data from §2.3, we find that only 46.3% of the greedily-decoded GPT-3-generated explanations for CommonsenseQA and 31.5% for NLI are rated acceptable **by 3/3 annotators**.⁸

Inspired by work in other generation tasks (Holtzman et al., 2020; Massarelli et al., 2020; Holtzman et al., 2021), we hypothesize that equally or more informative explanations can be generated by sampling stochastically. We sample 4 additional generations from GPT-3 for each instance to complement the greedy generation. We crowdsource 3 acceptability annotations for each new explanation.

As expected, sampled explanations exhibit lower 3/3 acceptability than greedy explanations (25.1% for CommonsenseQA; 11.3% for SNLI). However, this results in a surprisingly higher proportion of instances that have *at least one* acceptable explanation in the set of 5. The greedy explanation was judged to be 3/3 acceptable in 46.3% of instances for CommonsenseQA and 31.5% for NLI; this increases to 79.5% and 51.2%, respectively, when sampled explanations are included.⁹

4 Improving Explanation Generation with Acceptability Filtering

The challenge of overgeneration is that GPT-3 alone cannot discern *which* of its stochastic samples

are acceptable. Inspired by West et al. (2021), we explore training a supervised filter on the collected labels. Our key intuition is that by re-framing the role of annotators from explanation authors to binary judges, we can alleviate the need to collect a large-scale explanations dataset—the result is a simpler, cheaper, and easier crowdsourcing setup to administer (§4.1). Namely, we can (1) aggregate ratings over multiple annotators to produce more reliable labels, (2) use numerical metrics of annotator agreement to remove annotators contributing noisy data, and (3) collect annotations more quickly and cheaply than asking annotators to hand-write explanations. Moreover, we find that the filter can be trained with a relatively small amount of binary human judgments (§4.2). We demonstrate that the trained model is not simply taking advantage of surface-level features (§4.4). Figure 1 presents an overview of our pipeline.

4.1 Acceptability Annotations

We generate train and validation sets by repeating the procedure of generating 1 greedy and 4 sampled explanations for 991 and 1K instances, respectively, of the CommonsenseQA and SNLI training sets. Combining these with the annotated test sets from previous experiments results in a dataset of 1241/1250 instances in a 72/8/20% train/val/test ratio for each task. We again collect 3 binary acceptability ratings for each instance, resulting in ~6200 instance-explanation pairs and ~19K individual annotations per task. Table 13 contains statistics. To ensure that models trained on these corpora do not overfit to specific annotators (Geva et al., 2019), we collect an additional set of judgments for the test set of SNLI from a group of annotators who did not participate in any of our previous annotation tasks (“Test2”). Figure 9 and Figure 10 show the user interface.¹⁰

While we evaluate at test-time with the schema that only instances that 3/3 annotators deem acceptable are considered acceptable, preliminary experiments show that treating both 2/3 and 3/3 agreement instances as acceptable during training performs best on the 3/3 evaluation criterion at test-time.¹¹ We also train a variant where we randomly select one annotation from the three as the gold label (“without human agreement”).

⁸In §4, we show that these are not upper-bounds caused by intrinsic subjectivity, and that they can be improved upon.

⁹Appendix D provides similar results for the 2/3 acceptability threshold.

¹⁰Krippendorff’s α for all acceptability annotations is 0.34 for CommonsenseQA and 0.39 for SNLI (see Table 14).

¹¹Our results don’t significantly change if a 2/3 cutoff is used at test time instead: Appendix F contains the results.

4.2 Acceptability Filter

Concretely, given the problem instance, the gold label, and a generated explanation, the acceptability filter predicts whether the explanation is acceptable. We fine-tune two sequence-to-sequence architectures, T5-Large (Raffel et al., 2020) and T0-3B (Sanh et al., 2022). Each model is trained 5x with different random seeds. Further training details are given in Appendix E.

Baselines. We train an **explanation-only** baseline, which receives as input only the explanation; similar baselines have been proposed for NLI (Poliak et al., 2018; Gururangan et al., 2018). These models represent the hypothesis that annotator ratings can be reconstructed with only surface features of the explanation candidates, e.g., grammaticality. We also consider a **negative log-likelihood (NLL)** baseline, which uses GPT-3’s estimated probability as the acceptability classification score. This is a slightly more competitive baseline than greedy; greedy usually (but not always) produces the highest-likelihood explanation.¹²

4.3 Evaluation

We consider three evaluation settings. The first is **instance-level** (“select-1”), where the system returns 1 explanation selected from the set of 5 for each instance. We return the explanation with the highest model-estimated probability and report instance-level accuracy, i.e., the % of instances for which a gold acceptable explanation is selected.

We also evaluate at the **explanation-level**, where we treat each explanation independently and compute metrics over the full dataset. This aligns with the binary classification training of the models (cross-entropy on the explanation labels) and is suited for the setting in which we want to return *all* of the acceptable explanations per instance. In this setting, we report **average precision (AP)**, an estimate of area under the precision-recall curve.

Finally, we perform an absolute human evaluation (§2.3) on the subset of instances where the filter model does *not* select the greedy explanation as the best, i.e., comparing “select-1” performance to a greedy baseline on the instances where it differs. We additionally re-perform the head-to-head comparison of Table 5, replacing the greedy GPT-3 explanations with those selected by the filter.

¹²According to GPT-3, a sampled explanation from the set of 5 has a lower NLL than greedy for only 2.8% and 3.6% of instances, respectively, of our Com.QA and SNLI test sets.

	“Select-1” Acc@3/3		Expl.-level AP@3/3	
	Dev	Test	Dev	Test
Random	26.8 _{3.2}	30.6 _{2.1}	27.4 _{1.1}	31.6 _{0.4}
Constant	—	—	27.9	31.8
NLL	41.8	52.0	42.4	45.6
T5-L Expl.-only	40.2 _{3.9}	49.8 _{1.1}	43.5 _{1.5}	50.0 _{1.9}
T0-3B Expl.-only	42.6 _{1.4}	47.3 _{1.6}	41.1 _{2.0}	54.1 _{1.7}
T5-L w/o HA	46.6 _{2.3}	55.4 _{3.2}	47.0 _{2.3}	56.9 _{3.6}
T5-L	46.4 _{2.9}	55.4 _{2.1}	45.1 _{3.4}	58.3 _{1.5}
T0-3B w/o HA	48.4 _{2.0}	57.4 _{2.8}	44.5 _{2.3}	59.8 _{2.7}
T0-3B	48.6 _{0.9}	59.9 _{1.1}	49.7 _{1.6}	64.0 _{1.5}
Oracle U.B.	78.0	82.0	100.0	100.0

Table 7: Results for acceptability classifiers trained on CommonsenseQA. Subscripts indicate standard error over models trained with 5 different random seeds. “w/o HA” = without human agreement. “Oracle U.B” indicates upper bound based on dataset properties (§3).

	“Select-1” Acc@3/3			Explanation-level AP@3/3		
	Dev	Test	Test2	Dev	Test	Test2
Random	15.2 _{0.7}	14.7 _{0.1}	13.6 _{0.3}	15.0 _{0.6}	14.4 _{0.3}	13.8 _{0.2}
Constant	—	—	—	15.2	14.5	13.7
NLL	33.0	32.0	31.6	29.9	32.7	28.5
Expl.-only	30.2 _{0.8}	30.9 _{2.1}	27.8 _{1.8}	30.6 _{2.5}	30.6 _{1.3}	25.9 _{2.3}
w/o HA	38.2 _{1.9}	38.5 _{1.8}	36.2 _{1.4}	49.3 _{5.3}	48.5 _{3.3}	52.8 _{5.3}
Full	37.8 _{0.8}	38.7 _{0.7}	35.0 _{1.2}	46.8 _{3.6}	47.6 _{3.5}	49.5 _{4.8}
Oracle U.B.	51.0	52.4	46.4	100.0	100.0	100.0

Table 8: Results for SNLI explanation acceptability; all model results are on T0-3B. See Table 7’s caption.

4.4 Results

Classifier performance is given in Tables 7-8.

Effect of model size. On CommonsenseQA, T0-3B outperforms T5-Large by ~2-4% select-1 accuracy and ~5-6% explanation-level AP across splits. We use T0-3B in subsequent experiments.

NLL baseline vs. full model. For both tasks on both validation and test sets, T0-3B outperforms the NLL baseline substantially. On CommonsenseQA, we observe a 7-8% gain in instance-level accuracy, and a gain of 18% explanation-level AP on the test set. This provides strong evidence that the supervised model is able to incorporate binary human feedback to predict acceptable explanations at a level much higher than GPT-3 achieves on its own. We present examples where “select-1” predictions differ between NLL and our filter model in Table 10 and Table 16.

Explanation only vs. full model. Our results suggest that our models are leveraging feature interactions between the instance and explanation to make their predictions: without instance-level context, the explanation-only baselines are on average more than 5 points worse across metrics. Though

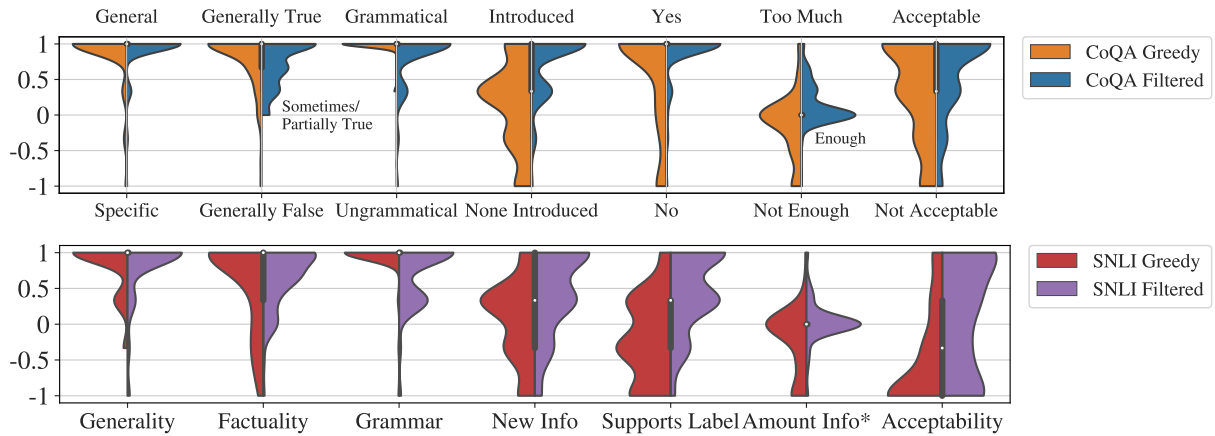


Figure 3: Absolute evaluation results in the “select-1” setting for the instances where our best-performing filter model does *not* select the greedy explanation (156 instances for CommonsenseQA (top); 91 for NLI (bottom)). See caption of Figure 2 and the Appendix-Table 20 for more details and statistical significance results.

Dataset	Preferred Explanation (%)			α
	Gold-Standard	Tie	GPT-3	
CoS-E	2.9	3.6	93.5	0.25
ECQA	19.9	10.0	70.1	0.05
e-SNLI	49.9	6.9	43.2	0.14

Table 9: A replication of Table 5, except the GPT-3 explanations are now the top-1 of our filter system.

they under-perform significantly relative to the full model, explanation-only baselines do fare surprisingly well, indicating that shallow features like factuality and grammaticality may represent latent factors in human acceptability judgments.

The effect of multiple training annotations. In some cases, performance improves if the training instances are labeled with the consensus of three annotators (vs. the singularly annotated case “w/o HA”), though the effects are inconsistent. In most cases, using consensus labels results in reduced variance across random seeds. However, the gains may not outweigh the 3x annotations required.

Our model doesn’t overfit to specific annotators. The performance of our model when evaluated on the NLI test set labeled by separate annotators (“Test2”) is comparable to the original test set (instance-level accuracy drops a few points, but explanation-level AP slightly rises).

Our model improves generated explanations along desirable traits. We present our absolute human evaluation for greedy vs. filtered explanations from GPT-3 in Figure 3— for both tasks, explanations filtered by our model more readily

SNLI

Premise: An officer in a black uniform and hat stands to the left of a large structure with other officers in the background.

Hypothesis: An officer enjoys coffee in a shop.

Label: contradiction

NLL-Predicted Explanation: An officer in a black uniform and hat is not necessarily an officer enjoying coffee in a shop.

Filter-Predicted Explanation: An officer in a structure standing to one side is not the same as enjoying coffee in a shop.

CommonsenseQA

Question: Where would there be an auditorium with only a single person speaking?

Answer choices: theater, park, university campus, crowd, or lights

NLL-Predicted Explanation: An auditorium is a large room used for lectures, and a single person speaking is likely to be a lecture.

Filter-Predicted Explanation: On university or university-like campuses, the auditoriums are often used for lectures or other University events, where a single person might be speaking.

Table 10: Randomly-selected instances that our filter model predicts differently than NLL at the “select-1” task and got **correct**, but NLL got incorrect.

introduce new information, support the label, and contain at least enough information for both tasks (in addition to being more acceptable). Interestingly, greedy explanations still prevail in surface-level features (grammaticality and, in the case of CommonsenseQA, factuality; differences are statistically significant with low p , see Table 20).¹³

We additionally find in our head-to-head study

¹³Krippendorff’s α for these experiments is 0.32 for CommonsenseQA and 0.33 for SNLI.

(Table 9) that, compared to Table 5, using filtered GPT-3 explanations instead of greedy increases the preference for GPT-3 explanations by 15-24% for both CommonsenseQA datasets. We do not see an increase in the SNLI case, which may be due to the fact that fewer GPT-3 explanations change after filtering (36.4%, compared to 62.4% for CommonsenseQA), and GPT-3 explanations for SNLI tend to be less acceptable overall, resulting in a lower upper-bound oracle of instances where an acceptable explanation can be selected (§3).

In summary. We have demonstrated the effectiveness of modeling binary crowd judgements of acceptability as a means to select candidates from GPT-3 which are deemed acceptable with unanimous agreement. For the method that does not leverage human agreement, this is done with only ~5k binary annotations. We additionally demonstrate that our filtered explanations improve upon greedy generations in fine-grained categories that probe their topical relevance and meaningful content. The gap between our best model and the upper-bound oracle indicates that there is still substantial room for improvement in both task settings (but especially for SNLI). Future work may investigate sampling more explanations, or incorporating other sources of supervision signal.

5 Related Work

Free-text explanation generation. The earliest neural free-text explanation models did so for computer vision applications (Hendricks et al., 2016; Park et al., 2018; Kim et al., 2018) and NLI (Camburu et al., 2018). These methods relied on supervised datasets to train the explanation generator. Others have proposed to generate explanations or clarifications to improve task performance in a supervised (Rajani et al., 2019; Lampinen et al., 2022) or unsupervised (Shwartz et al., 2020) manner. Yordanov et al. (2021) study transfer learning between datasets for few-shot generation.

Latcinnik and Berant (2020) proposed a method to generate free-text explanations supervised only on task signal, and Brahman et al. (2021) used sources of weak supervision to generate explanations for defeasible inference. Paranjape et al. (2021) design hand-crafted templates which they use with mask-infilling to produce contrastive explanations from pretrained language models.

Concurrent work (Marasović et al., 2021) studies the effect of prompt format and model size on

crowdworker judgements of prompted explanation plausibility. They find that GPT-3 Davinci outperforms other smaller pretrained models, but that crowdworkers find these explanations less plausible than those from the datasets, aligning with our first experimental result (Table 3). We perform a more in-depth study of the fine-grained criteria comprising human acceptability, and demonstrate that with higher-quality prompts and filtering, GPT-3’s performance can be significantly improved.

Supervising on human preferences. Prior and concurrent work has used binary judgements from crowdworkers to fit models to human preferences for non-XAI tasks such as summarization (Ziegler et al., 2020; Stiennon et al., 2020), creating commonsense knowledge bases (West et al., 2021), and building natural language inference datasets (Liu et al., 2022). Unlike these works, we apply human preference modeling to increase the human acceptability of model-generated free-text explanations. West et al. (2021) demonstrate that GPT-3 + a supervised acceptability filter can generate a high-quality causal knowledge graph: in addition to their work being conducted in a different domain, our success conditions and evaluation metrics differ because we *must* produce a prediction for each instance (whereas they can simply discard bad generations).

6 Conclusion

We demonstrate GPT-3’s capacity to generate free-text explanations for NLP task instances in a few-shot setting. We further improve this capability via an overgenerate + filter approach, where the filter is trained on supervision from human acceptability ratings. We hope our results can guide future work on free-text explanations via neural or neuro-symbolic systems (Brahman et al., 2021; Majumder et al., 2021; Saha et al., 2021). Future work may also further investigate the benefits of counterfactual explanations.

While human rationales for decision making are not necessarily the same as model rationales, the goal behind modeling human acceptability is often to build trust with a human user. This trust may or may not be warranted (Jacovi et al., 2021); future work would be well-suited to further investigate generated explanations for incorrect label predictions such as in Appendix C, which could mislead end users.

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7 Ethics & Broader Impacts

All datasets used in this work are public, and we plan to release the machine-generated explanations and annotations we collected. We do not collect any personal information from our human participants.

Models that produce explanations in the means used in our experimental protocol (i.e., by conditioning on the gold labels) have the possibility to cause humans to place unwarranted trust in an AI system. This line of research is complementary to works investigating the faithfulness of model-generated free-text explanations (Hase et al., 2020; Wiegrefe et al., 2021). We demonstrate in Appendix C that GPT-3’s explanations lack reliability because the model can explain answer choices that were not its prediction equally well. This may be due in part to the fact that decoding algorithms for generating predictions from language models are sub-optimal (e.g., Zhao et al., 2021; Holtzman et al., 2021) and GPT-3 may have factual knowledge stored in its parameters about other answer choices that allow it to provide reasonably acceptable explanations. Until this phenomenon is better understood, we do not condone using GPT-3-generated explanations in real-world deployment.

Lastly, our model of human acceptability is based on the aggregate judgements of participants from primarily Western, English-speaking countries working on crowdsourcing platforms. The subjective judgements of explanation acceptability may vary significantly among different population groups.

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A Prompt Construction

Following [Perez et al. \(2021\)](#), we avoid prompt tuning on the full training and development sets of the datasets studied, in order to ensure that our methods represent a true few-shot setting. To develop the initial prompt design, we experimented with no more than 10 different layouts in the GPT-3 Sandbox platform using 15 training examples on the CoS-E and e-SNLI datasets. For subsequent prompt design, we again used these 15 training examples for each dataset from which we sampled 6 prompts,

Let’s explain classification decisions.
question: When remembering a tragedy in the past, what do many people feel?
pain, depression, knowing, knowledge, or nostalgia?
depression
why? Remembering a past tragedy can resurface feelings that arose in response to that tragedy. Because tragedies are not positive events, it’s possible that sadness and depression could arise from remembering it.

question: What do people do sometimes when they find a good deal?
race cars, murder each other, believe in god, fight each other, or fear death?
fight each other
why? Malls sometimes have sales, e.g., on black friday, when they offer good deals; however, the items are sometimes in limited supply, which can cause altercations between folks, each trying to buy the same item.

question: What does someone who has a greed for energy do?
buy food, lie, get, cause businesses to grow, or win?
buy food
why? When consumed, food provides energy and satisfies the greed for it.

question: Immediately after peeing, a person’s bladder is what?
full, empty, filled, stretchable, or collapsed?
empty
why?

Table 11: Example of a prompt with 3 training examples for CommonsenseQA: presented are the **question** and **answer choices**, the **gold labels**, and the **explanations** (written by us) that act as input to GPT-3 (in practice, we use 8-24 examples per prompt). The text generated by the model acts as the free-text explanation. In this case, the model greedily auto-completes (given 8 examples): “After peeing, the bladder is empty.”

along with a fixed 100-example “development set” randomly drawn from the training set. We preserve the “few-shot” approach by using a maximum of these same 115 instances to develop our prompting methods. For these 115 examples, the authors of this paper manually wrote high-quality explanations to be used as prompt examples (Table 12). As presented in Table 2, we found that structuring SNLI as a question-answering task achieved the best performance, similarly to Zhao et al. (2021). We provide an example of our SNLI prompt in Table 2 and CommonsenseQA in Table 11.

In-context learning methods have been shown to have high variance based on hyperparameters including example order, number of examples given, and which examples are given (Jiang et al., 2020; Zhao et al., 2021; Lu et al., 2022). While these

values have not been standardized, two prominent papers, Schick and Schütze (2021b) and Brown et al. (2020), use 32 and 64 prompt examples, respectively. Due to the 2049-token limit of the OpenAI GPT-3 API and the fact that the addition of explanations elongates each prompt instance, we find the maximum number of examples the API can accommodate is 24 for CoS-E, e-SNLI, and our handwritten explanations and 16 for ECQA.

The focus of this work is not on finding the optimal prompt, but on developing a general strategy for few-shot explanation generation that could be successful when no additional (large) validation set for tuning is available. Therefore, to provide as robust of an expected performance estimate as possible, we do not tune the additional hyperparameters, instead sampling them to approximate performance.¹⁴ Namely, while prior work uses one fixed prompt for all instances and varies the random seed, we approximate the same expected performance by sampling a new set of prompts for each instance. We also sample the number of prompts for each instance (and shuffle their order) from the values {8, 16, 24} for CommonsenseQA experiments, {8, 16} for experiments using ECQA explanations, and {12, 18, 24} for SNLI experiments (to maintain label balance). To overcome label bias in prompt ordering, for tasks with distinct answer choices per instance (CommonsenseQA), we shuffle the answer choices. For tasks with fixed answer choices (SNLI), we sample an equal number of prompt instances for each label (so number of prompt instances is a multiple of 3).

Table 12 shows a few non-cherry-picked examples of our handwritten explanations used as prompts relative to the datasets.

B Crowdsourcing Details

We discuss shared details of the study designs in §B.1. We discuss the head-to-head interface in §B.2, the absolute interface in §B.3, and the acceptability interface in §B.4. Finally, we present details on quality control and payment in §B.5 and annotator statistics in §B.6.

B.1 Shared Interface Details

For all three human subjects study designs designs, we show the user the input instance (e.g.,

¹⁴Perez et al. (2021) show that performing cross-validation or tuning via maximum description length over a small validation set does not significantly outperform random sampling of these values.

SNLI (Bowman et al., 2015)
Premise: A person on a horse jumps over a broken down airplane. Hypothesis: A person is training his horse for a competition. Label: neutral Our Explanation: While it is possible that jumping a horse over an obstacle is part of a training routine for a competition, it is also possible that the horse ride is being done for pleasure, not necessarily for a competition (sp). e-SNLI Explanation: the person is not necessarily training his horse
Premise: Children smiling and waving at camera Hypothesis: There are children present Label: entailment Our Explanation: Since the children are part of the event of smiling at the camera, they are present at the event under discussion. e-SNLI Explanation: The children must be present to see them smiling and waving.
CommonsenseQA (Talmor et al., 2019)
Question: A cat can't talk, but a cat can what? Answer choices: sleep all day, <u>meow</u>, shed fur, see king, live many years Our Explanation: A cat can meow as a way to vocalize. CoS-E Explanation: the cat is a small carnivorous mammal ECQA Explanation: A cat can meow but cannot see the king. Meowing is how a cat communicates and not by sleeping all day, shedding fur or by living many years.
Question: "There are 10 apples on an apple tree. Three fall off. Now there are X apples." What is this an example of? Answer choices: park, coloring book, garden center, <u>math problem</u>, gravity Our Explanation: A math problem is usually posed as a question that requires some operation such as subtraction or addition to answer. CoS-E Explanation: webmath is designed to help you solve ECQA Explanation: Math problem is an arithmetical problem of addition, subtraction, multiplication or division. So "There are 10 apples on an apple tree. Three fall off. Now there are X apples." is a math problem. All the other options aren't problems to be examples of the given question.

Table 12: Examples of explanations used as prompts from various sources, including our handwritten explanations. Correct answers for CommonsenseQA are underlined.

premise+hypothesis) and the gold label in addition to the explanation(s). We explain our motivation for using the gold label as a methodological control in §2.

For a similar reason, we do not show the other incorrect label choices to the user, which is particu-

larly of note for the CommonsenseQA task which has different answer choices per instance. Some instances in CommonsenseQA have multiple correct or very similar answer choices, due to noise in the dataset and the fact that the wrong answer choices were deliberately collected to make the task challenging. We (the authors) again found we struggled to accurately judge explanation quality when we disagreed with the selected answer choice or found multiple answer choices to be correct. To remove this possible confounder, we instruct participants to pretend the gold label is correct even if they disagree with it, and make this easier by hiding the other answer choices. This may result in a slight bias in judgements against the ECQA dataset due to its unique counterfactual nature, though our goal was not to study the benefits and downsides of counterfactual explanations in this work.

B.2 Head-to-Head Interface Details

We show the user the task input and gold label, and ask them to select which of two explanations best explains the answer. We instruct workers to consider the gold label to be correct even if they disagree with it (CommonsenseQA instances can be subjective) and to ignore minor grammar and spelling mistakes such as improper upper-casing. Figures 5 and 6 show the evaluation interface.

B.3 Absolute Interface Details

Figures 7 and 8 show the absolute evaluation interface (minus the acceptability attribute, which is collected in a separate run of the study). Our interface is inspired by prior work from psychology and the social sciences (Leake, 1991; Gopnik, 1998; Lombrozo, 2007; Zemla et al., 2017; Chiyah Garcia et al., 2018; Clinciu et al., 2021; Sulik et al., 2021). We iterated over 3-4 versions of the questions and UI design until we had optimized agreement rates as much as possible. Our resulting two-part evaluation consists of 7 questions:

Part 1: Context-Independent Evaluation We first assess the explanation in isolation, i.e., these questions are presented to the user without revealing the question/context that the explanation is attempting to address:

1. *How factual is this statement?* (generally false, sometimes or partially true, generally true, or need more information to judge). This question is designed to test both generality (can the explanation's truthfulness be ascertained or is more in-

formation needed?) and factuality, which aligns with “compatibility with receiver’s existing beliefs” and that the best explanation is the “most likely” explanation from the receiver/user’s perspective (Lombrozo, 2007; Zemla et al., 2017; Sulik et al., 2021). Generality is coded based on whether a truthfulness answer is selected (considered to be general) or whether the “need more information to judge” choice is selected (considered not to be general).

2. *Is this statement grammatical?* (yes or no) This question is designed to test for clarity, aligning with characteristics such as coherence (Lei et al., 2016) and human-likeness and understandability (Ehsan et al., 2019).

Part 2: Context-Dependent Evaluation We next show the user the question (premise and hypothesis for SNLI) and gold answer that the explanation was conditioned on. We then ask:

1. *Does the explanation provide new facts, information or reasoning not stated in the question and answer?* (yes or no) In our preliminary experiments, we found some explanations simply restate the question declaratively with the answer filled in. This question addresses the distinction between “validity” and “utility” (Leake, 1991): an explanation can be valid (i.e., a restatement of the question with the answer filled-in might be correct), but not useful; utility is defined by whether an explanation “satisfies an explainer’s need for information”. And while utility is best understood in the context of real-world applications (Lai et al., 2020), we nonetheless aim to identify vacuous explanations that do not provide new information.
2. *Is the new information relevant to justifying the answer?* (yes or no) New information, if provided, “should be compatible with our existing beliefs, and consistent with the evidence and with itself” (Zemla et al., 2017). This question is designed to test whether the information provided supports the label. The specific interpretation of “relevance” is purposefully left to the annotator.¹⁵
3. *How much information does the explanation have to justify the answer?* (not enough, enough, or too much) This question is designed to test

¹⁵This decision is inspired by prior work in psychology, which finds that explanations are only good “to the extent that people find [them] satisfying” (Gopnik, 1998; Sulik et al., 2021).

the extent to which the provided novel information is *adequate* or *sufficient* (Kim et al., 2016; Lei et al., 2016; Ehsan et al., 2019).¹⁶

4. *Is the explanation acceptable?* (yes or no) The final question is designed to assess annotators’ overall judgement of the explanation as a whole.

We only ask Question 2 if the answer to Question 1 is “yes” and Question 3 if the answer to Question 2 is yes, because they regard the new facts, information, or reasoning. We found that most prior work tends to lump added-value, relevance, and adequacy judgements into one “informativeness” judgement (Clinciu et al., 2021), which we felt was too coarse to allow for meaningful error analysis.

B.4 Acceptability Interface Details

Figures 9 and 10 show the binary acceptability interface used to collect training and test data for the overgeneration filter model.

Spearman’s rank-order correlation coefficients (Table 6) are computed using scipy (Virtanen et al., 2020) on the 250 test explanations from the 5 data sources in Figure 2. Each instance is annotated by 3 annotators for a total of 3750 datapoints (some criteria are only evaluated conditionally, resulting in less total annotations— see Appendix B.3). Statistical significance is computed using the built-in two-sided non-parametric test.

B.5 Quality Control and Payment

We use Amazon Mechanical Turk (AMT), and calculate pay on a rate of \$15/hour. Every few batches, we check to ensure that the median time taken per-annotator amounts to approximately this pay rate. While annotators do tend to speed up the more HITs we released, first-round median times were approximately 30 seconds per head-to-head evaluation HIT (thus paid at \$0.12 each), 1 minute per absolute evaluation HIT (thus paid at \$0.25 each), and 35-39 seconds per acceptability HIT (5 explanations; paid at \$0.20 each).

We require annotators to be located in either Australia, Canada, New Zealand, the United Kingdom, or the United States, as a proxy for English competency.¹⁷ We require a past HIT approval rate of >

¹⁶In practice, we do not find Turkers use the “too much information” option often, except in the case of ECQA dataset explanations. We included the option because succinctness is an oft-cited explanatory virtue (Lombrozo, 2007; Zemla et al., 2017; Chiyah Garcia et al., 2018).

¹⁷We realize this is a broad assumption and likely sub-optimal. However, colleagues have found that broadening

98% and > 5000 HITs approved. We do not allow annotators to participate who were previously on a block list from our past AMT studies.

Annotators must complete a qualifying exam in order to participate in the main annotation tasks. The qualifying exam consists of 3 HITs in the same format as the main absolute evaluation task for CommonsenseQA. We pay \$2.25 for the qualifying exam. There are 9-18 questions in total (3-6 questions per HIT), some of which are only answerable conditioned on previous answers. A user who answers “no” to question 3, for example, will not be asked to answer questions 4 and 5. Given the challenging and sometimes ambiguous nature of some of the questions, for the first run of the qualification exam, we manually awarded qualifications by inspecting the annotators’ answers. Scores for the first run compared to our answers (out of 17 annotators attempting) ranged from 5 to 14 out of 18. The median accuracy was 11 out of 18, and we find that awarding the qualification to those with scores at or above the median aligns closely with our manual inspection. We thus use this score to assign qualifications in future iterations.

Because it is necessary that annotators understand the task before they can evaluate explanation quality (Wiegrefe and Marasović, 2021), for tasks that are more difficult, i.e., NLI, we additionally require annotators to pass (score of 7 or above) a task-specific qualification exam with 8 questions, paid at \$1.25.

In order to track quality throughout evaluation, we compute inter-annotator agreement using Krippendorff’s α and use a hidden built-in Javascript function to compute time per HIT spent. If any annotator completed the tasks in an unreasonably low time, or removing their annotations substantially improves Krippendorff’s α , we remove their annotations and re-annotate their instances. We additionally ensure that each experiment has a substantial number of distinct crowdworkers to mitigate individual annotator bias, reporting this as well as the mean and median number of HITs completed by each in Table 17.

B.6 Statistics

The number of distinct crowd annotators and the median and mean number of HITs completed for each experiment can be found in Table 17. More

the geographical requirements often still leads to >90% of annotators in the US or Canada, due to AMT’s pay structure being optimal in these countries.

Dataset	Split	# Instances by Agreement				Total
		0/3	1/3	2/3	3/3	
Com.QA	Train	932	1078	1194	1296	4500
	Dev	105	91	132	127	455
	Test	298	227	328	397	1250
SNLI	Train	2372	805	621	702	4500
	Dev	272	87	65	76	500
	Test	678	225	166	181	1250
	Test2	666	234	179	171	1250

Table 13: Statistics of our acceptability annotations.

detailed breakdowns of inter-annotator agreement for some experiments are in Tables 14 and 15.

C Absolute Evaluation by Label Accuracy

Can GPT-3 produce convincing explanations even for instances it cannot predict correctly? This has implications for model-generated explanations being “right for the right reasons”. To produce label predictions, we follow the same prompt format as in Tables 2 and 11, removing the WHY? token and the gold explanations so that the model generates a label prediction instead. GPT-3 achieves 50.8% accuracy on CommonsenseQA compared to a 20% random baseline, and 46% accuracy on SNLI compared to a 33.33% random baseline.¹⁸

Figure 4 presents absolute evaluation results broken down by whether GPT-3 correctly predicts the instance label. The results show little variation between the correctly-predicted and incorrectly-predicted groups in most attributes. This indicates that GPT-3 explanations are not faithful enough to use in real-world applications in their current form.

D 2/3 Acceptability Statistics

When we treat explanations rated by at least 2/3 annotators as “acceptable”, for CommonsenseQA, 77.9% of greedily-decoded explanations are acceptable; for SNLI, 51.0%. 50.5% of sampled explanations are acceptable; for SNLI, 23.5%. Out of the set of 5 (1 greedy + 4 stochastic), 97.7% of CommonsenseQA instances have *at least one* acceptable explanation, and 79.5% of SNLI.

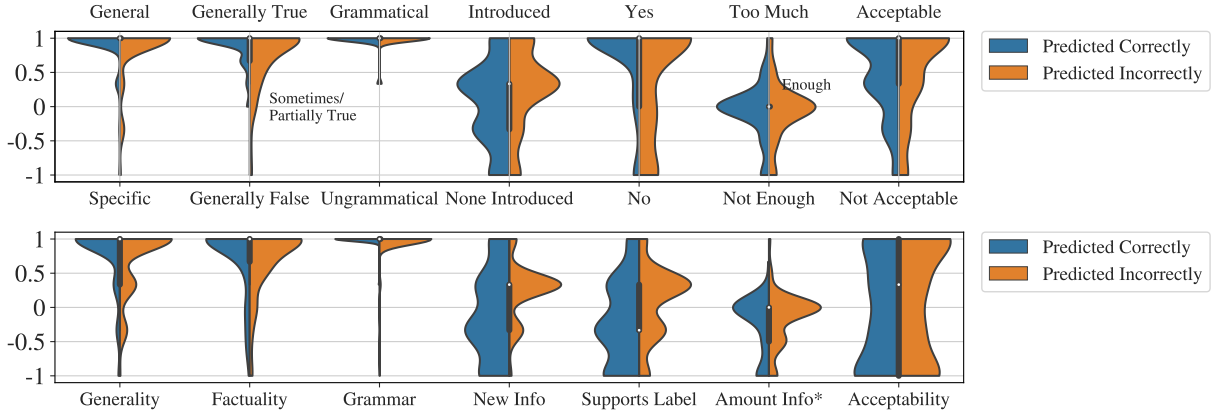


Figure 4: Absolute evaluation results for explanations generated by GPT-3 based on whether GPT-3 predicted the instance label correctly or not. CommonsenseQA (top) and SNLI (bottom). See caption of Figure 2 and the Appendix-Table 19 for more details.

E Filter Model Details

We split the 4,955 distinct annotated explanations for CommonsenseQA (5000 for SNLI) into a train/dev set of 4500/455 (4500/500 for SNLI), where all 5 explanations for a given instance are placed in the same set to avoid leakage. We present statistics on the label distribution in Table 13. Along with the metric settings reported in the paper (“select-1” and explanation-level), we computed a metric that is instance-level but considers all explanations by computing metrics over the 5 explanations of an instance and then averaging across instances, finding in practice that the results are highly similar to the explanation-level evaluation.

We use Huggingface Datasets (Lhoest et al., 2021) and Huggingface Transformers (Wolf et al., 2020) for implementation. We format inputs to the models as follows, where `expl` is one of the five explanations and the `gold_label` is either 0 (not acceptable) or 1 (acceptable):

```
if explanation_only:
    input_string = (f"explanation: {expl}.
    Is this explanation good or bad?")
else:
    input_string = (
        "{question} answer: {gold_label}. "
        "explanation: {expl}. "
        "Is this explanation good or bad?")
```

The T5-Large model is trained using a learning rate of $1E - 4$ with linear decay, a batch size of 64, and default values for Adam (Kingma and Ba, 2015), gradient clipping, and dropout. We train for

Dataset	Split	Krippendorff's α
CommonsenseQA	Training + Validation	0.32
	Test	0.40
SNLI	Training + Validation	0.51
	Test	0.50
	Test2	0.47

Table 14: Inter-annotator agreement for acceptability AMT studies.

a maximum 200 epochs, performing early stopping on the validation loss with a patience of 10 epochs. For T0-3B, we train with a batch size of 50. We use AdaFactor (Shazeer and Stern, 2018) with a linear warmup of 500 steps. We conduct an initial hyperparameter sweep over learning rate, considering $1E - 5, 5E - 05, 5E - 06$. The learning rate that achieves the best validation loss for the full-information model and the explanation-only model is $1E - 5$, which we use for all training experiments.

F Additional Filter Results

In the main experiments, at evaluation time, we labelled an explanation as acceptable if 3/3 annotators agreed on it. Here, we report results if this threshold is relaxed to 2/3. Overall, the results are comparable: T0-3B outperforms the baselines according to both select-1 accuracy and AP (see Table 21 and Table 22).

¹⁸Low SNLI performance aligns with previous findings that GPT-3 struggles with sentence-comparison tasks (Brown et al., 2020; Zhao et al., 2021).

AMT Study	Dataset	Generality	Factuality	Grammar	New Info	Supports Label	Amount Info	Acceptability	Aggregate
GPT-3 Greedy	Com.QA	0.37	0.32	-0.01	0.09	0.45	0.21	0.28	0.38
GPT-3 Greedy	SNLI	0.25	0.57	0.39	-0.01	0.04	0.17	0.52	0.40
Dataset	CoS-E	0.71	0.38	0.36	0.42	0.68	0.08	0.25	0.59
Dataset	ECQA	0.01	0.21	0.30	0.00	0.03	0.25	0.04	0.20
Dataset	e-SNLI	0.37	0.23	0.37	-0.14	-0.12	0.04	0.15	0.19
GPT-3 Filtered	Com.QA	0.25	0.18	0.25	0.11	0.27	0.15	0.30	0.32
GPT-3 Filtered	SNLI	0.41	0.19	0.07	0.17	0.08	0.13	0.50	0.33

Table 15: Inter-annotator agreement for absolute-comparison AMT studies, using Krippendorff’s α computed on an interval scale from -1 to 1. The aggregate score is computed by treating the annotations along each attribute as separate instances and computing agreement across the entire set.

SNLI (Bowman et al., 2015)

Premise: There are two kilted men, one of them older and is holding bagpipes with the other one with a drum.

Hypothesis: Two kiled (sp) men hold bagpipes

Label: contradiction

NLL-Predicted Explanation: The two kilted men are not holding bagpipes, they are holding a bagpipe and a drum.

Filter-Predicted Explanation: Just because there are kilted men does not necessarily mean that they are holding bagpipes. This can be seen from the older kilted man is holding the bagpipes rather than the kilder (sp) one.

CommonsenseQA (Talmor et al., 2019)

Question: The hardcovers were especially tall, so he removed a shelf on the what to make room?

Answer choices: hold alcohol, grocery store, bookcase, nightstand, or chest of drawers

NLL-Predicted Explanation: Bookcases are tall, and so if they are full of hardcover books, they might need to be removed to make room for other things.

Filter-Predicted Explanation: People keep books on bookshelves, and bookcases are very commonly used to hold books. Since hardcover books can be pretty tall, it takes up extra vertical space which would require removal of an upper shelf of the bookcase.

Table 16: Randomly-selected instances that our filter model predicts differently than NLL at the “select-1” task, and got **incorrect** but NLL got correct.

AMT Study	Task/Dataset	# Annotators	Median # HITs (Mean)
GPT-3 Greedy w/ Dataset Prompts vs. Dataset	Com.QA/CoS-E	16	31.5 (46.9)
	Com.QA/ECQA	13	35 (57.7)
	e-SNLI	12	39 (62.5)
GPT-3 Greedy: Author-written vs. Dataset Prompts	Com.QA/CoS-E	7	84 (107.1)
	Com.QA/ECQA	13	49 (57.7)
	e-SNLI	8	43.5 (93.8)
GPT-3 Greedy w/ Author-written Prompts vs. Dataset	Com.QA/CoS-E	8	90 (93.8)
	Com.QA/ECQA	17	27 (44.1)
	e-SNLI	8	93 (93.8)
GPT-3 Greedy (Absolute)	Com.QA	13	51 (57.7)
	SNLI	12	14 (62.5)
Dataset (Absolute)	CoS-E	14	58 (53.6)
	ECQA	19	7 (39.5)
	e-SNLI	13	16 (57.7)
Acceptability (Training and Validation Data)	Com.QA (2973 HITs)	34	70 (87.4)
	SNLI (3000 HITs)	14	128.5 (214.3)
Acceptability (Test Data)	Com.QA	17	32 (44.1)
	SNLI	11	26 (68.1)
	SNLI (Test2)	7	65 (107.1)
	CoS-E	13	48 (57.7)
	ECQA	16	38.5 (46.9)
	e-SNLI	9	60 (83.3)
GPT-3 Filtered (Absolute)	Com.QA (468 HITs)	10	44.5 (46.8)
	SNLI (273 HITs)	6	53 (45.5)

Table 17: Total # of annotators and mean # HITs completed per-annotator for each AMT study (out of 750 total # HITs unless otherwise specified = 3 annotators for each of 250 test instances).

Set of Test Explanations	Generality	Factuality	Grammar	New Info	Supp. Label	Amt. Info	Accept.
GPT-3 Greedy for Com.QA	0.9_{0.4} [‡]	0.8_{0.4} (247) [‡]	1.0_{0.1} [‡]	0.1 _{0.6}	0.5_{0.7} (217) [‡]	-0.1_{0.4} (186) [‡]	0.5_{0.6} [‡]
CoS-E	-0.2 _{0.9}	0.5 _{0.5} (131)	-0.3 _{0.7}	0.1 _{0.8}	-0.3 _{0.9} (190)	-0.5 _{0.5} (78)	-0.9 _{0.4}
GPT-3 Greedy for Com.QA	0.9_{0.4} [∨]	0.8_{0.4} (247) [‡]	1.0_{0.1} [‡]	0.1 _{0.6}	0.5 _{0.7} (217)	-0.1 _{0.4} (186)	0.5_{0.6} [‡]
ECQA	0.8 _{0.4}	0.6 _{0.4} (249)	0.1 _{0.7}	0.6_{0.5} [‡]	0.7_{0.5} (247) [^]	0.5_{0.5} (239) [‡]	0.1 _{0.6}
GPT-3 Greedy for SNLI	0.7_{0.5} [^]	0.7 _{0.5} (246)	1.0_{0.2} [‡]	0.1 _{0.6}	-0.1 _{0.6} [*]	-0.2 _{0.4} (203)	0.0 _{0.8}
e-SNLI	0.6 _{0.6}	0.8 _{0.4} (236)	0.9 _{0.4}	0.2_{0.5} [∨]	0.2_{0.5} ^{*‡}	-0.1_{0.4} (238) [^]	0.7_{0.4} [‡]

Table 18: Statistics from the graphs plotted in Figure 2. Mean $\pm$ standard error presented; numbers in parenthesis indicate the number of datapoints, if not 250. *For SNLI, we modified the evaluation framework such that “Supp. Label” was always answered instead of being conditioned on “New Info”. Statistical significance results using a one-sided Wilcoxon signed-rank test at p -values of $\ddagger = 0.00001$, $\dagger = 0.0001$, $\vee = 0.01$, $\wedge = 0.1$ indicates that the median difference between the marked score distribution and the unmarked score distribution is greater than 0.

Set of Test Explanations	Generality	Factuality	Grammar	New Info	Supp. Label	Amt. Info	Accept.
Com.QA Pred. Correctly	0.9_{0.3} (127) [∨]	0.9_{0.3} (126) [∨]	1.0 _{0.1} (127)	0.0 _{0.6} (127)	0.7_{0.6} (108) [‡]	-0.1 _{0.4} (98)	0.5 _{0.6} (127)
Com.QA Pred. Incorrectly	0.8 _{0.4} (123)	0.7 _{0.5} (121)	1.0 _{0.1} (123)	0.3_{0.6} (123) [∨]	0.3 _{0.8} (109)	-0.1 _{0.4} (88)	0.5 _{0.7} (123)
SNLI Pred. Correctly	0.8_{0.5} (115) [^]	0.7 _{0.5} (112)	1.0_{0.1} (115) [^]	-0.1 _{0.7} (115)	-0.2 _{0.6} (115)	-0.3 _{0.4} (83)	-0.1 _{0.8} (115)
SNLI Pred. Incorrectly	0.7 _{0.5} (135)	0.7 _{0.5} (134)	1.0 _{0.2} (135)	0.2_{0.4} (135) [‡]	0.1_{0.5} (135) [∨]	-0.2_{0.4} (120) [^]	0.1_{0.8} (135) [^]

Table 19: Statistics from the graphs of GPT-3 greedy explanations plotted in Figure 4. See the caption of Table 18 for further details.

Set of Test Explanations	Generality	Factuality	Grammar	New Info	Supp. Label	Amt. Info	Accept.
GPT-3 Greedy for Com.QA	0.9 _{0.4} (156)	0.8_{0.4} (153) [∨]	1.0_{0.1} (156) [‡]	0.1 _{0.6} (156)	0.5 _{0.7} (135)	-0.1 _{0.5} (117)	0.3 _{0.7} (156)
GPT-3 Filtered for Com.QA	0.9_{0.3} (156) [^]	0.7 _{0.3} (155)	0.8 _{0.4} (156)	0.7_{0.4} (156) [‡]	0.9_{0.4} (154) [‡]	0.2_{0.3} (152) [‡]	0.6_{0.6} (156) [∨]
GPT-3 Greedy for SNLI	0.8 _{0.4} (91)	0.6 _{0.6} (91)	0.9_{0.3} (91) [‡]	0.0 _{0.6} (91)	-0.2 _{0.6} [*] (91)	-0.2 _{0.5} (66)	-0.5 _{0.7} (91)
GPT-3 Filtered for SNLI	0.8 _{0.5} (91)	0.7 _{0.4} (88)	0.7 _{0.4} (91)	0.5_{0.6} (91) [‡]	0.5_{0.5} [*] (91) [‡]	0.0_{0.3} (89) [‡]	0.1_{0.8} (91) [‡]

Table 20: Statistics from the graphs plotted in Figure 3. See the caption of Table 18 for further details.

↓Model/Split→	“Select-1” Acc@2/3		Explanation-level AP@2/3	
	Dev	Test	Dev	Test
Random	57.3 _{0.4}	57.9 _{0.4}	56.2 _{0.5}	58.0 _{0.9}
Constant	—	—	56.9	58.0
NLL	79.1	79.6	77.5	75.0
T0-3B Expl.-only	77.1 _{3.5}	75.8 _{1.2}	75.6 _{2.0}	77.3 _{1.4}
T0-3B	86.6_{0.9}	85.8_{0.7}	85.6_{0.5}	87.0_{0.8}
Oracle Upper-Bound	97.8	97.6	100.0	100.0

Table 21: Results for acceptability classifiers trained on CommonsenseQA, with “acceptability” defined as: “at least 2/3 annotators labelled as acceptable.” Subscripts indicate standard error over models trained with 5 different random seeds. The oracle upper bound is based on dataset properties (§3).

↓Model/Split→	“Select-1” Acc@2/3			Explanation-level AP@2/3		
	Dev	Test	Test2	Dev	Test	Test2
Random	28.2 _{0.5}	27.8 _{0.2}	28.0 _{0.1}	28.1 _{0.9}	27.6 _{0.3}	28.3 _{0.6}
Constant	—	—	—	28.2	27.8	28.0
NLL	51.0	51.2	50.4	47.7	47.5	46.1
T0-3B Expl.-only	47.0 _{1.0}	50.5 _{2.1}	50.6 _{2.8}	48.9 _{1.4}	45.2 _{1.5}	44.9 _{2.1}
T0-3B	57.8_{1.9}	60.3_{1.5}	59.2_{2.3}	66.7_{3.3}	64.7_{3.3}	67.1_{3.6}
Oracle Upper-Bound	76.0	81.2	77.6	100.0	100.0	100.0

Table 22: Results for acceptability classifiers trained on SNLI with “acceptability” defined as: “at least 2/3 annotators labelled as acceptable.” Subscripts indicate standard error over models trained with 5 different random seeds. The oracle upper bound is based on dataset properties (§3).

Instructions (click to expand/collapse)

Thanks for participating in this HIT!

You will read multiple-choice questions about daily life. For each question, an answer has been selected and 2 explanations for the answer have been given. Your task is to select the explanation that best explains the answer.

Each HIT contains 3 parts:

Question	A question such as "If a lantern is not for sale, where is it likely to be?"
Answer	A selected answer, such as "house" (may or may not be correct).
Explanations	Two explanations for the Answer .

A couple of notes:

- You may disagree with the **Answer**, but you should pretend it is correct when selecting the best **Explanation**.
- You can ignore minor grammar and spelling mistakes. Some sentences will be lowercased incorrectly; please ignore this.
- If you find the **Explanations** similarly good (or bad), select the one you think is the best and then select *"I voted, but this one is close because the explanations are similar."*
- If you find the **Explanations** equally good (or bad) and cannot choose, select *"Both equally good."* or *"Both equally bad."*

Examples (click to expand/collapse)

Question: \${question}

Answer: \${gold_label}

Which explanation better explains the answer?

\${explanation_1}

\${explanation_2}

☐ Both equally good.
☐ Both equally bad.
☐ I voted, but this one is close because the explanations are similar.

(Optional) Please let us know if anything was unclear, if you experienced any issues, or if you have any other feedback for us.

Submit

Figure 5: An overview of the user interface of our head-to-head comparison AMT studies for CommonsenseQA. The top shows the instructions and the bottom the actual task. The Examples tab is collapsed here; shown in full in Figure 6.

Examples (click to expand/collapse)

Example #1

Question: If a lantern is not for sale, where is it likely to be?

Answer: house

Which explanation better explains the answer?

A house has lanterns.

people keep furniture, such as lanterns, at their houses for personal use in their daily living.

The better explanation is **people keep furniture, such as lanterns, at their houses for personal use in their daily living.** It explains why "house" answers the part of the question that specifies "not for sale", which the first choice does not explain.

Example #2

Question: "there are 10 apples on an apple tree. three fall off. now there are x apples." what is this an example of?

Answer: math problem

Which explanation better explains the answer?

The example is of a question that involves performing subtraction.

webmath is designed to help you solve

The better explanation is **The example is of a question that involves performing subtraction.** The second explanation is nonsensical.

Example #3

Question: Where is the most likely place for more than one stadium?

Answer: large city

Which explanation better explains the answer?

a stadium requires a lot of space.

a good place for more than one stadium is a place with a lot of people.

The better explanation is **a good place for more than one stadium is a place with a lot of people.** It explains why a large city would have multiple stadiums. The first explanation is factually correct, but does not explain, for example, why a large city would have more space than a rural area.

Figure 6: The Examples tab given in the user interface of our head-to-head comparison AMT studies for CommonsenseQA. The full interface is shown in Figure 5.

Instructions (click to expand/collapse)

Thanks for participating in this HIT!

You will read multiple-choice questions about daily life. For each question, an answer has been selected and a statement explaining the answer has been given.

Part 1: Please read the **statement** and answer questions related to its factuality and grammaticality (2 questions).

Part 2: Please read the **instance** that the statement explains and answer questions about the explanation quality (1-3 questions).

An instance contains 3 parts:

Question	A question such as "If a lantern is not for sale, where is it likely to be?"
Answer	A selected answer, such as "house" (may or may not be correct).
Explanation	The statement from Part 1, which explains the Answer .

[Important!] You may disagree with the **Answer, but you should pretend it is correct when judging the **Explanation**.**

[Important!] Some sentences will be lowercased incorrectly; please ignore this.

[Important!] You will not be able to change your answers to Part 1 after moving on to Part 2.

Main Example (click to expand/collapse)

More Examples (click to expand/collapse)

Part 1: Please read the below statement and answer the following questions:

\$_{predicted_explanation}

1) How factual is this statement?

2) Is this statement grammatical?

Submit Part 1

Figure 7: An overview of the user interface template of our absolute comparison AMT studies for CommonsenseQA. The top shows the instructions and the bottom the actual task. Only part 1 of the task is shown here (part 2 appears once part 1 is submitted). The Main Example and More Examples tabs illustrating both parts 1 and 2 are collapsed here; see Figure 8.

Main Example (click to expand/collapse)

Part 1: Please read the below statement and answer the following questions:

people keep furniture, such as lanterns, at their houses for personal use in their daily living.

1) How factual is this statement?

Generally False Sometimes or Partially True Generally True Need More Information to Judge

This statement is **Generally True**.

2) Is this statement grammatical?

No Yes

The correct answer is **Yes**.

Part 2: Please read the instance that the statement explains and answer the following questions:

Question: If a lantern is not for sale, where is it likely to be?

Answer: house

Explanation: people keep furniture, such as lanterns, at their houses for personal use in their daily living.

3) Does the **Explanation** provide new facts, information, or reasoning not stated in the **Question** and **Answer**?

Examples

No Yes

The correct answer is **Yes**, because the explanation introduces the fact that lanterns are a type of furniture, and that people do not usually keep furniture in their house for selling.

4) Is the new information **relevant** to justifying the **Answer**? Examples

No Yes

The correct answer is **Yes**.

5) How much information does the **Explanation** have to justify the **Answer**? Examples

Not Enough Enough Too Much

The correct answer is **Enough**.

Figure 8: The Main Example given in the user interface template of our absolute comparison AMT studies for CommonsenseQA. This format follows the actual task layout. The full interface is shown in Figure 7.

Instructions (click to expand/collapse)

Thanks for participating in this HIT!

You will read questions about daily life. For each question, an answer has been selected and 5 explanations have been given.

Your task is to judge whether each explanation is acceptable.

Each HIT contains 3 parts:

Question	A question such as "If a lantern is not for sale, where is it likely to be?"
Answer	A selected answer, such as "house" (may or may not be correct).
Explanations	Five explanations for the Answer .

A couple of notes:

- You may disagree with the **Answer**, but you should pretend it is correct when judging the **Explanations**.
- Some sentences have minor grammar and spelling mistakes, don't end on a period, or cut off partway through because they are long. Please ignore these, and judge on the part of the sentence that is presented.
- New!** If at least part of the explanation is acceptable and sufficient (even if some other part is nonsensical or rambling), please mark it as acceptable.

Examples (New!) (click to expand/collapse)

Question: \${question}

Answer: \${gold_label}

Which explanations are acceptable? Select all that you believe adequately explain the answer (this may be none).

☐ \${explanation_1}

☐ \${explanation_2}

☐ \${explanation_3}

☐ \${explanation_4}

☐ \${explanation_5}

(Optional) Please let us know if anything was unclear, if you experienced any issues, or if you have any other feedback for us.

Submit

Figure 9: An overview of the user interface of our explanation acceptability AMT studies for CommonsenseQA. The top shows the instructions and the bottom the actual task. The "examples" tab is collapsed here; shown in full in Figure 10.

Example #1:

Question: A bald eagle flies over St. Paul, where is it?
Answer: minnesota

Which explanations are acceptable? Select all that you believe adequately explain the answer (this may be none).

- ☐ Bald eagles are native to the United States, but are rarer in Minnesota than in other states because Minnesota has a small supply of their primary foods (fish, mice, etc.).
- ☐ St. Paul is a city in Minnesota, so a bald eagle flying over it would likely live in Minnesota.
- ☐ St. Paul is in Minnesota.
- ☐ The question assumes that you've seen the bald eagles that live in Minnesota.
- ☐ Bald eagles live in northern North America, and can be found anywhere there. Given that one is flying over St. Paul, it must be either migrating, or it could be a photographer, making its appearance in the city unusual.

Possible answer:

Bald eagles are native to the United States, but are rarer in Minnesota than in other states because Minnesota has a small supply of their primary foods (fish, mice, etc.).

St. Paul is a city in Minnesota, so a bald eagle flying over it would likely live in Minnesota.

St. Paul is in Minnesota.

The question assumes that you've seen the bald eagles that live in Minnesota.

Bald eagles live in northern North America, and can be found anywhere there. Given that one is flying over St. Paul, it must be either migrating, or it could be a photographer, making its appearance in the city unusual.

One possible answer is that only the second and third explanations are acceptable. The first, fourth, and fifth explanations do not provide an explanation for the fact that St. Paul is in Minnesota.

Another possible answer is that only the third explanation is acceptable (since the second explanation incorrectly states that the bald eagle *lives* in Minnesota). However, we prefer that you err on the side of being forgiving if part of the explanation is correct.

Example #2:

Question: Even if was clean and chlorinated, shallow water wouldn't make for much summer fun in your backyard what?
Answer: swimming pool

Which explanations are acceptable? Select all that you believe adequately explain the answer (this may be none).

- ☐ It would be illegal to use and non-sanitary to use a lake or other body of water as a swimming pool.
- ☐ Pools seem to be the kind of shallow water that people would swim in for fun.
- ☐ Swimming pools get deeper than puddles or ponds.
- ☐ Pools are the only body of water that contain both swimming-friendly water and shallow water.
- ☐ Swimming pools are often used for summer fun.

Possible answer:

It would be illegal to use and non-sanitary to use a lake or other body of water as a swimming pool.

Pools seem to be the kind of shallow water that people would swim in for fun.

Swimming pools get deeper than puddles or ponds.

Pools are the only body of water that contain both swimming-friendly water and shallow water.

Swimming pools are often used for summer fun.

The question implies that the answer must be a body of water that can get deep, and must be able to be used for summer fun.

One possible answer is that only the fifth explanation is acceptable. The first, second, and fourth explanations do not explain that pools can get deep or that pools can be used for summer fun. The fifth explanation explains at least one of these facts.

Another possible answer is that both the third and the fifth explanations are acceptable, since the third explanation indirectly states that pools can get deep (although it is somewhat debatable if they are deeper than ponds).

Figure 10: The examples given in the user interface of our explanation acceptability AMT studies for CommonsenseQA. The full interface is shown in Figure 9.